Solving a Higher Order (Quartic) Equation by Considering $u = x + \frac{1}{x}$

Solve $x^4 - 8x^3 + 17x^2 - 8x + 1 = 0$.
Given that all the roots are: $x = \frac{3+\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2}, \frac{5+\sqrt{21}}{2}, \frac{5-\sqrt{21}}{2}$.

Proof:

$$\begin{aligned} x^4 - 8x^3 + 17x^2 - 8x + 1 &= 0 \\ x^2 - 8x + 17 - \frac{8}{x} + \frac{1}{x^2} &= 0 \\ \left(x^2 + \frac{1}{x^2}\right) - 8\left(x + \frac{1}{x}\right) + 17 &= 0 \\ \left[\left(x + \frac{1}{x}\right)^2 - 2\right] - 8\left(x + \frac{1}{x}\right) + 17 &= 0 \\ \left(x + \frac{1}{x}\right)^2 - 8\left(x + \frac{1}{x}\right) + 15 &= 0 \end{aligned}$$

Let $u = x + \frac{1}{x}$. Then, we have

$$\begin{aligned} u^2 - 8u + 15 = 0 \\ u = 3 \quad \text{or} \quad 5 \end{aligned}$$

Hence, we have

$$\begin{aligned} x + \frac{1}{x} &= 3 & \text{or} && x + \frac{1}{x} &= 5 \\ x^2 - 3x + 1 &= 0 & \text{or} && x^2 - 5x + 1 &= 0 \\ x &= \frac{3+\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2} & \text{or} && x &= \frac{5+\sqrt{21}}{2}, \frac{5-\sqrt{21}}{2} \end{aligned}$$